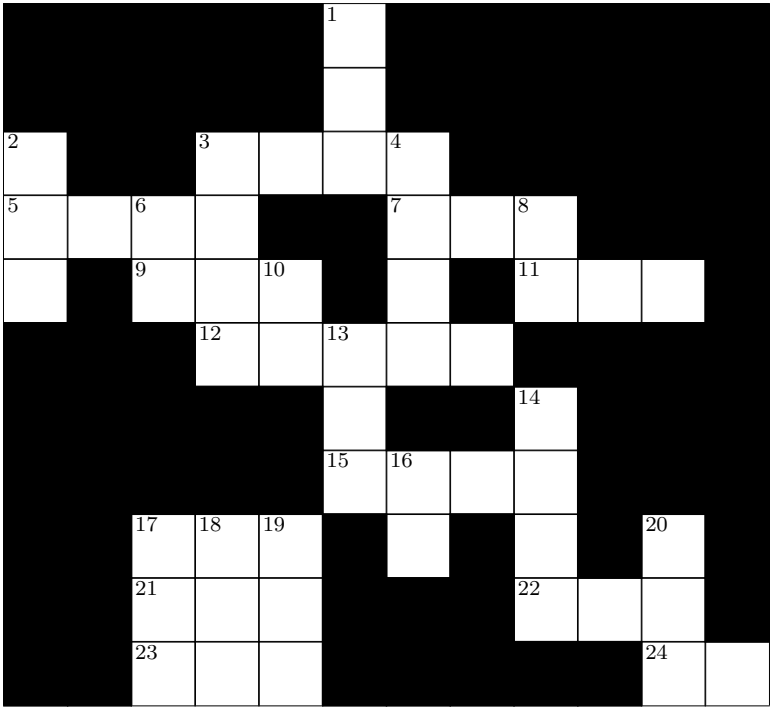
MASTER COPY

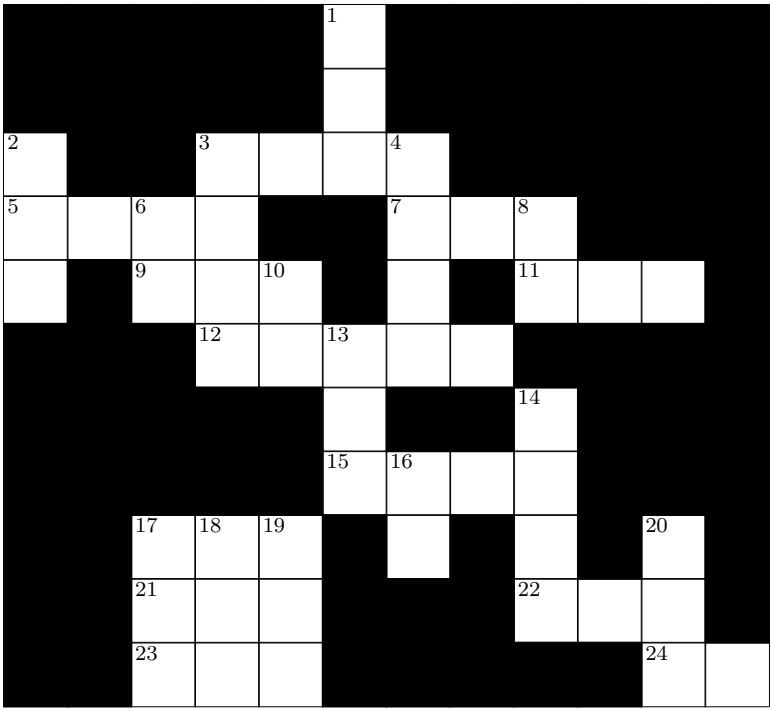


ALL NUMBERS IN THIS COPY ARE CORRECT.

ONLY GRADERS MAY WRITE ON THIS COPY.

Team Name:

TICKS:



Happy numbers

A non-negative integer is happy if, by repeatedly replacing the number with the sum of the squares its digits, we eventually reach 1. Example:

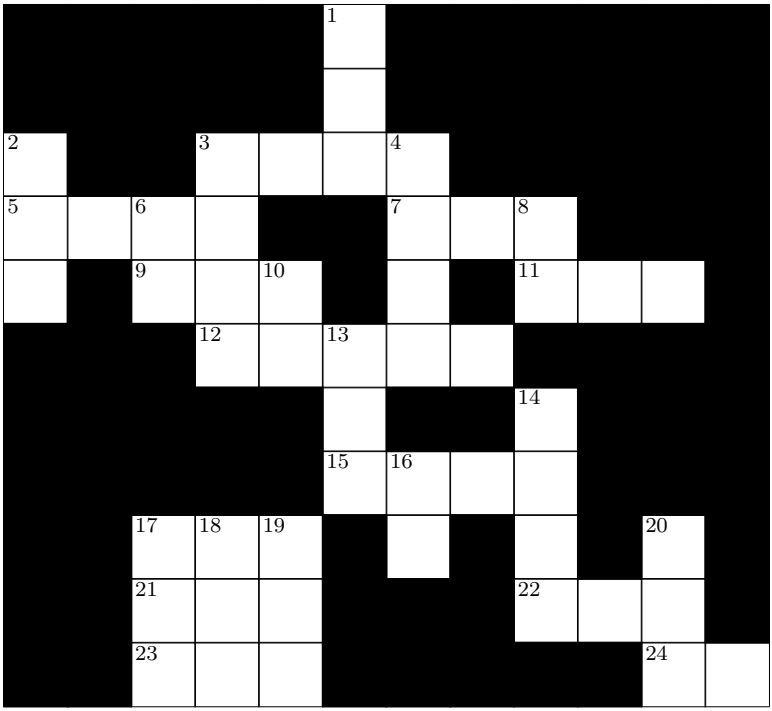
- $86 \to 8^2 + 6^2 = 100 \to 1^2 + 0^2 + 0^2 = 1$, thus 86 is happy.

A non-negative integer which is not happy is unhappy.

Palindromic numbers

A non-negative integer is palindromic if it reads the same forwards and backwards. Examples:

- 212 is palindromic – it reads 212 backwards.
- 69 is not palindromic – it reads 96 backwards.



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